INSTITUTIONS, TRANSACTION COSTS AND ECONOMIC GROWTH

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This essay develops a theoretical framework which explores the historical obstacles to economic growth. These obstacles are examined in the context of the political/economic institutional framework of economies in history and consequent transaction costs that determine economic performance and growth. The essay concludes with specific suggestions for the study of economic growth.

I

The objective of this essay is twofold: to develop a theoretical framework which focuses on the historical obstacles to economic growth; and to suggest the implications of the framework for the study of poverty, equity, and growth in developing countries.

I begin with one of the most cited but still misunderstood essays of our time. It is twenty-five years since Ronald Coase published "The Problem of Social Cost" (1960), but the impact of that essay has still not really penetrated the economics profession or the development literature. Coase pointed out that the neoclassical model, which has served as the basis of economic reasoning for most scholars in the Western world, holds only under the severely restrictive assumption of zero transaction costs; but that with positive transaction costs, institutions matter. There are no institutions in the neoclasical world; and indeed in such a world, growth is not a problem, its rate being simply a function of the number of children people have and the rate of saving.[1]

Ever since, Adam Smith economists have recognized that gains from trade are the key to the wealth of nations. Specialization and division of labor have made possible the improved productivity that arises from technological change, better resource allocation, and specialized production, the key underlying features of modern economies. What economists have not realized until recently is that exchange is not costless. Economists still misunderstand the costs involved in exchange, regarding exchange as costless (as the standard neoclassicists assume) or unproductive (as in the classical notion of unproductive labor), or contending that such costs exist but are passive, and therefore not important, or are neutral with respect to their consequences for economies.

* The author wishes to thank Lee Benham, Elisabeth Case, Ken Koford, Donald McCloskey, Kenneth Shepsle, John Wallis and Barry Weingast for their comments on an earlier draft of this essay. The original version of this paper was prepared for a conference on "The Political Economy of Poverty, Equity, and Growth" sponsored by the World Bank in Cancun, Mexico, December, 1985.

1. See North [1981] chapter I for an elaboration of the implications of the neoclassical model for economic growth.

Economic Inquiry
Vol. XXV, July 1987, 419–428

In fact the costs of transacting are the key to the performance of economies. There have always been gains from trade, as classical international trade theory has taught; but so too have there been obstacles to realizing these gains. If transport costs were the only obstacle, then we would observe through history an inverse relationship between transport costs, on the one hand, and trade and exchange and the well-being of societies on the other. But recall that as early as the Roman Empire of the first two centuries A.D. trade was possible over a vast area, even with the transport costs of the time; and that after the end of the Roman Empire trade declined and probably the well-being of societies and individual groups declined as well. It was not that transport costs had risen, but that the costs of transacting had risen with the disappearance of a unified political system and effectively enforced rules and laws over a large area.

From the evidence of history, let us turn to the economies of the world today and observe the enormous disparity between the rich countries of the western world and the poor countries of the third world. It is not transport costs but the costs of transacting that are the key obstacles that prevent economies and societies from realizing well-being. We can understand why when we examine analytically the costs of transacting in different situations.

I begin with a simple model of personal exchange. In personal exchange individuals either engage in repeat dealings with others or otherwise have a great deal of personal knowledge about the attributes, characteristics, and features of each other. As a result, the measured transaction costs in such a society are very low because of the dense social network of interaction. Cheating, shirking, opportunism, all features that underlie modern industrial organization theory, are limited or indeed absent, because they simply do not pay. Norms of behavior are seldom written down. Formal contracting does not exist, and there are few formal specific rules. However, while measured transaction costs in such societies are low (although unmeasured costs of societal cooperation in tribal societies may indeed be high), their production costs are high, because specialization and division of labor are limited by the extent of the market defined by personal exchange.

At the other extreme from personal exchange is a world of specialized interdependence in which the well-being of individuals depends upon a complex structure characterized by individual specialization and hence exchange ties extended both in time and space. A pure model of this world of impersonal exchange is one in which goods and services or the performance of agents is characterized by many valued attributes, in which exchange takes place over time, in which there are not repeat dealings. Under these forms of exchange, the costs of transacting can be be high, because there are problems of being able to measure the attributes of what is exchanged and problems of enforcement of the terms of exchange; in consequence there are gains to be realized by engaging in cheating, shirking, opportunism, etc.[2] In order to prevent such

2. See Lancaster [1966], Becker [1965], Cheung [1983], North [1981], and Barzel [1982] for an elaboration of the underlying argument.

activity, elaborate institutional structures must be devised that constrain the participants and so minimize the costly aspects I have just described. As a result, in modern Western societies we have devised formal contracts, bonding of participants, guarantees, brand names, elaborate monitoring systems, and effective enforcement mechanisms. In short, we have well-specified and well-enforced property rights. The result of all this is that resources devoted to transacting are large (although small per transaction) but the productivity associated with the gains from trade is even greater; and high rates of growth and development have characterized Western societies. Of course these institutions in turn depend on a much more complex institutional structure that makes possible the specification and enforcement of property rights, which in turn allow transactions to occur and productivity gains from modern technology to be realized.

Increasing specialization and division of labor necessitate the development of institutional structures that permit individuals to take actions that involve complex relationships with other individuals, which require both personal knowledge and time. The evolution of more complex social frameworks will not occur if such institutional structures cannot reduce the associated uncertainties. The essential institutional reliability means that we can have confidence in outcomes increasingly remote from our personal knowledge as the network of interdependence caused by the growth of specialization widens.[3]

The institutional requirements that are necessary in order to be able to realize the productivity gains associated with the model of impersonal exchange outlined above entail the development of efficient product and factor markets and a medium of exchange with reliable features. The establishment and enforcement of such a set of property rights will then allow individuals in highly complex interdependent situations to have confidence in their dealings with strangers. This is only possible as the result, first, of the development of a third party to exchanges, namely government, which specifies property rights and enforces contracts; and second, of the existence of norms of behavior to constrain the parties in interaction, which will permit exchange where high measurement costs, even with third party enforcement, pose problems with respect to opportunism, cheating, and so forth.

But why isn't it automatic to develop more and more complex institutions that will enable us to handle more complex interdependence? Indeed, much of the literature of game theory and the stories of institutional development imply that the progress of primitive societies to the status of modern Western societies should be automatic and unilinear. The answer is quite clear. The breakdown of personal exchange is not just the breakdown of a dense communication network but is the breakdown of communities of common ideologies and of a common set of rules that all believe in. The rise of impersonal rules and contracts means the rise of the state and with it an unequal distribution of coercive power. This provides the opportunity for individuals with superior coercive power to enforce the rules to their advantage, regardless of

3. Ronald Heiner [1983] explores these issues in an interesting essay.

their effects on efficiency. That is, rules will be devised and enforced on behalf of the interests of the politically advantaged, but they will not necessarily lower the costs of transacting in total.

In fact, one of the most evident lessons from history is that political systems have an inherent tendency to produce inefficient property rights which result in stagnation or decline. There are two basic reasons for this result. First, the revenue that can be raised by rulers may be greater with an inefficient structure of property rights that can, however, be effectively monitored and therefore taxed, than with an efficient structure of property rights that has high monitoring and collection costs.[4] Second, rulers can seldom afford efficient property rights, since they would offend many of their constituents and hence become more insecure. That is, even when rulers wish to promulgate rules on the basis of their efficiency consequences, survival will dictate a different course of action, because efficient rules would offend powerful interest groups in the polity.[5]

Increasing specialization also produces another costly consequence, which is that the common ideologies and norms of behavior characterizing tribal communities disappear as individuals have different experiences, different perceptions of reality, and therefore conflicting views about how a society should be structured and what their own interests are in that society.

Finally, time enters into the story. We observe that the creation of new rules does not in itself immediately produce new stable forms of behavior. Rules and norms of behavior are deeply ingrained in human beings, and security is associated with a given code of conduct and behavior, which will only gradually change. Thus the key to understanding institutions, and indeed the dilemma of institutional change, is to recognize that for individuals, perceptions about what constitutes codes of conduct and rules that they can live with will only gradually evolve over a lifetime. Institutional analysis is at base the study not simply of the rules of the game but of the individual responses to such rules. While the rules may change overnight, individual responses will be much more complex and slow to adapt. Changes in rules require the evolution of norms, conventions, and informal codes.

The issues can be summarized in more formal terms. Modern economic growth results from the development of institutions that permit an economy to realize the gains from specialization and division of labor associated with the sophisticated technology that has developed in the Western world in the last several centuries.[6]

There are four major variables involved in the costs of exchange which fundamentally influence the rate of growth of economies. The first is the cost of measuring—measuring the valuable attributes of the goods and services being exchanged or measuring the performance of agents. Measurement at-

4. The granting of monopolies to guilds in Colbert's France produced this result.

5. For an elaboration of these points, see "A Neo-Classical Model of the State," in North [1981] and North [1984].

6. These issues are dealt with at greater length in North [1984].

tempts to delineate precisely what the participants are exchanging or what performance characteristics are expected of agents. It is essential to the specification and enforcement of property rights and underlies all the costs of contracting. In a world of perfect and costless information, measurement would be unnecessary and there would be no transaction costs, since each party already would know precisely what was being exchanged or what was the performance of agents; and enforcement would be unnecessary, since both parties would costlessly know the outcome of disputes ahead of time. The immense resources that societies devote to organization and enforcement would be superfluous, and the Walrasian system would indeed provide a model for the economy. But because information is costly and imperfect at best, the other three variables in the cost of transacting become important.[7]

The second variable is the exchange process resulting from the size of the market. In personal exchange, kinship ties, friendship, and personal loyalty all play a part in constraining the behavior of participants, or repeat dealing and personal knowledge of the other participants constrain behavior. All of these factors reduce the need for costly specification and enforcement. A handshake suffices for even complicated exchange. Absent these factors, and the exchange process becomes much more costly. In impersonal exchange there is nothing to constrain the parties from taking advantage of each other if they can gain by doing so. Accordingly, contracts must be specified as precisely as possible and elaborate safeguards developed to enforce compliance.

Enforcement is the third variable. In a world of perfect enforcement there would be a neutral third party impartially evaluating disputes and awarding compensation to the party injured as a result of contract violation. In such a world, opportunism, shirking, and cheating would never pay. But such a world does not exist, because the costliness of measurement frequently makes it difficult to determine whether a contract has been violated (and by whom). Nevertheless, an impersonal body of law, courts, and the coercive power to enforce judgements are fundamental factors in permitting the complex contracting essential to a world of specialization and impersonal exchange.

The first three variables determining the costs of exchange arise as a consequence of an environment in which individuals maximize at every margin. If cheating pays, one cheats; if loafing on the job is possible, one loafs; if one could with impunity burn down a competitor, one would do so. In the neoclassical world, behavior is only constrained by rules and the effectiveness of their enforcement. Although the neoclassical assumption has been a powerful tool in analyzing market exchange, it is clearly incomplete. In the context of the high costs of measuring performance, complex social organization would be impossible were everyone to murder, cheat, steal, and shirk whenever he found it profitable. Honesty, integrity, living up to the spirit as well as the letter of the law do matter. But how much?

The neoclassical assumption is that preferences or tastes matter but are

7. Barzel [1982, 27–48] and North [1981].

constant. But ideological attitudes, the fourth variable, are not constant. The strength of ideology can be measured by the premium people are willing to incur rather than "free ride." Such a premium is the cost of one's convictions. The premium varies according to one's beliefs about the justice of the rules and the contractual arrangements of the society, and it can be influenced by education, propaganda, and symbols.[8] Appeals to justice and fairness do matter; otherwise we would be at a loss to explain a good deal of schooling as well as the immense investments made by politicians, employers, labor leaders, and others in trying to convince participants of the fairness or unfairness of a contractual arrangement. Our understanding of ideology is still sketchy, but one additional point about it appears to be significant. Specialization and division of labor produce divergent perceptions of reality and hence contrasting and conflicting views of the fairness and justice of institutional arrangements. In such a world the cost of contracting, *ceteris paribus*, increases because one cannot rely on common perceptions of fairness to enforce compliance of agreements.

These four variables have combined to produce an enormous variety of forms of economic organization, which can be subsumed, however, under three general types: personal exchange, impersonal exchange without third-party enforcement, and impersonal exchange with third-party enforcement. Let me elaborate upon each of these "ideal" types.

Personal exchange, characterizing small-scale production and local trade, has been the predominant form of organization throughout history. Cultural homogeneity (that is, a common set of values) and a lack of third-party enforcement (and indeed, little need for it) have been typical. Under these conditions transaction costs are low; but because specialization and division of labor are rudimentary, production costs are high.

Impersonal exchange without third-party enforcement covers an enormous variety of types of organization, spanning relatively large-scale retail (and wholesale) trading, long-distance and cross-cultural trade, and forms of organization in which one party has overwhelming coercive power relative to another. The *suq* or bazaar economies illustrate the first of these.[9] The *commenda* and variations upon it typify the second; serfdom and slavery the third.[10] In the case of the bazaar and *commenda* the lack of third-party enforcement leads to elaborate efforts to establish repeat dealings and clientization, to use kinship ties, and to set the trade within the context of rituals and religious precepts in order to constrain the participants. In the case of

8. This definition is not incompatible with other views about ideology in which it is a form of economizing on information costs and a signalling device. What I do wish to stress is that it is also concerned with normative views of the world.

9. Clifford Geertz, Hilda Geertz, and Lawrence Rosen [1979, 123–313]. Geertz makes clear that it is the high costs of information and lack of third party enforcement together with impersonal exchange that produces the characteristics of *suq* exchange.

10. It should be carefully noted that, because of the high costs of measurement and enforcement, the stronger party still must take into account the actions of the weaker party; hence the usefulness of this approach even with coerced agreements.

serfdom and slavery the state must be an interested party in enforcing the terms of exchange in favor of one party. The three forms of organization have in common high costs of transacting and limited gains from trade. Specialization and division of labor tend to be limited to the specific form of organization and a specific type of economic activity but not extended to the economy as a whole.

Impersonal exchange with third party enforcement is the essential requirement for sustained economic growth. A society that would achieve the productive benefits of great specialization can only do so with an elaborate structure of law and its enforcement. There are no cases of complex high income societies that do not have an elaborate structure of government. In such economies, impersonal exchange involves the high measurement costs of complex contracting necessary to realize the potential of the technology that comes from specialization. Neither self-enforcement by the parties themselves nor trust is a viable way of enforcing such contracts. It is not that ideology does not matter. It does; and as described above, immense resources are devoted to attempting to promulgate codes of conduct. But equally, the rate of return on opportunism, cheating, shirking, and so on also rises in such a context. A coercive third party is essential. One cannot have the productivity of an industrial society with political anarchy. But while such a state is a necessary condition for realizing the gains from trade, it obviously is not sufficient. With a state inevitably comes a struggle to control it in the interests of one of the parties. The state then becomes the vehicle by which the costs of transacting are raised to capture the gains that will accrue to any interested party that can control the specification and enforcement of property rights. The point has been the subject of an extensive literature that appears to have obscured the key point. If you want to realize the potential of modern technology you cannot do with the state, but you cannot do without it either.

II

Because modern technology is capital-intensive, characterized by large-scale production and continuous output, modern economies are necessarily complex and interdependent systems.[11] A large variety of economic institutions are necessary to facilitate such an economic structure in order to engage in contracting which 1) has reliability, 2) specifies the nature of the exchanges as precisely as possible, and 3) minimizes the potentiality of opportunism. The key features of the institutional structure of modern economies are contracting that permits continuous production and distribution and also the enforcement of contracts over long periods of time.

Underpinning this set of economic institutions must be a structure of political-judicial institutions that provide for stability of property rights. Stability implies that there not be radical changes in the price level and that property rights are imbedded in a structure of law far removed from arbitrary confis-

11. For an elaboration see North [1981] Chapter 12 or Chandler [1977].

cation or alteration. There must be a high degree of enforcement of property rights, which involves substantial resources devoted both to enforcement and to mitigating the principal-agent problem, so that agents who enforce the law are, to a substantial degree, impartial in their judgements. Finally, there must be a structure of law and its enforcement that gives parties confidence that over time contracts will be enforced.

Ideally, such a body of property rights are flexible, in the sense of allowing a variety of organizational forms for long-term and short-term contracts, thus encouraging the growth of spot markets, firms, and financial markets of all kinds that allow one to take advantage of this technology.

It is one thing to be able to describe the institutional structure of successful economies; it is something else altogether to be able to specify how one can realize such an economic-political system. Here the new political economy is still in its infancy. Most of the research to date explores the political structure of well-developed economies rather than the evolution of political economic systems that turn out to be successful. Still, our historical knowledge of the successful economies does gives us some clues. When we examine the Netherlands and England in the sixteenth, seventeenth, and eighteenth centuries, we observe the development of market-induced pressures to alter the political system in such a way as to embed rules and their enforcement in an impartial body of law as far removed as possible from arbitrary changes. The development of representative government in these two successful economies is consistent with this development, for representative government implies a body of law that is impersonal and not subject to arbitrary confiscation or alteration by rulers. Such a concept of the polity was envisioned by James Madison in Federalist Paper No. 10. But the evolution of a political economic system in the Anglo-Saxon world, as characterized by the development of English Parliament and the development of the American colonies and the early American nation, was a long and gradual process, one in which the interplay between the expanding markets and the development of political institutions took a number of centuries. The story that Madison portrayed and that other Federalist Papers also attempt to describe mirrored several centuries of gradual evolution of economic and political systems and of ideas about these systems. Such evolution cannot be replicated in a short period of time.

The application of a transactions cost framework to exploring the evolution of political structures does offer promise of yielding some interesting results. A key feature of the transaction cost framework, as applied to economic institutions, is that individuals will structure contracts in such a way as to minimize the dissipation of rent at all margins in contracting.[12] The same principles are behind Madison's Federalist Paper No. 10. Madison was attempting to devise a set of political insitutions that make it unprofitable for the participants to engage in rent dissipating activities and that encourage them to engage in productive activities. We are a long way from developing

12. See Barzel [1982].

such models, although recent studies such as Robert Bates' *Markets and States in Tropical Africa* (1981) suggest the lines for application of the new political economy in exploring political-economic interdependence in Third World countries.

III

How can we measure the efficiency of political and economic institutions? Traditional national income accounting conceals rather than reveals the critical features of institutional efficiency for three reasons: the cost of transacting may be so high that no production occurs, transactions costs are not separated out from other costs in the National Income Accounts, and some costs of transacting do not go through the market and therefore are not measured. Let me examine each in turn. The first problem is by far the most difficult to deal with. Increasing productivity by realizing the gains from specialization requires the development of an institutional framework to lower the costs of transacting (per exchange). Property rights that are unstable, that have perverse incentives (i.e., disincentives to undertake productive investment), or that cannot be enforced will be evidenced by a failure to develop the institutions of finance, credit, banking, marketing, insurance, etc., which would make possible transacting at a profitable production cost. The absence of such institutions when compared to their presence in successful developing countries would provide essential clues for exploring production failures as a result of the high cost of transacting.

The second problem is solvable and indeed, a preliminary study of measuring the transaction sector in the American economy from 1870 to 1970 has been undertaken in a study by North and Wallis in Volume 51 of the Income and Wealth Series.[13] It includes a workable definition of transaction costs that can be employed in other studies and a guide to the way the transaction sector has evolved in a successful economy. Furthermore, it highlights the increasing absorbtion of transaction costs inside firms, the growing productivity of wholesale and retail trade, and the relatively minor share of transacting undertaken by government (even though the government enforcement of property rights is a critical part of the story). Similar studies of other economies over time will allow us to develop generalizations about the evolving structure of economies and enable us to pinpoint problems of development. When we have such studies, we can draw inferences about an efficient development pattern associated with an evolving structure of the transaction sector. We can also then develop some meaningful comparisons with relatively inefficient transaction sector economies—for example, the bazaar economies of many underdeveloped Third World countries in which the inefficiency of the wholesale-retail trade structure leads to enormous resources devoted to relatively inefficient transacting.[14]

13. North and Wallis [1986].
14. For a more detailed study of the costs of transacting in bazaar economies, see Geertz [1979].

The third problem is no different than the traditional problems posed in national income accounting, in which substantial resources do not go through the marketplace and therefore measurements of the transaction sector do not capture them. In particular, transactions costs that do not go through the marketplace reflect the high costs of searching where information is not efficiently distributed and substantial costs in undertaking economic activity in compliance with rules and regulations. In particular, costs of queueing in societies that do not allow for the growth of market sectors will not be reflected in the transaction sector. Detailed case studies of the costs of queueing, of the time (and hence costs) involved in getting permits to do business, of cutting through red tape, of bribing the necessary officials, etc., are essential to understand the nature and importance of nonmarket transaction costs.

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